CO2 - AT1 - Analytical problem Solving

Evaluation of postfix Expression using Stack.

Given postfix Expression

2354 +9 -

A postfix expression is an expression in which the Operator is written after the operands, A Stack is used to evaluate the expression.

Because it follows the LIFO (Last In, First Out) principal.

Symbol	Operation	Stack contents
2	push 2	2
3	push 3	2, 3
*	$2 \times 3 = 6$	6
5	push 5	6, 5
4	push 4	6, 5, 4
*	$5 \times 4 = 20$	6, 20
+	$6 + 20 = 26$	26
9	push 9	26, 9
-	$26 - 9 = 17$	17

FINAL RESULT = 17

Algorithm to Evaluate any postfix Expression using a stack

Algorithm: postfix_Evaluation

Step 1: Start

Step 2: Create an empty stack

Step 3: Scan the postfix expression from Left to right

Step 4: if the ~~second~~ Scanned symbol is an operand

Push the operand into the stack

Step 5: if the Scanned symbol is an operator

Step 6: Repeat steps 4 and 5 Until the entire expression is Scanned

Step 7: The final element remaining in the Stack is the result

Step 8: Stop

Why Stack is more Suitable than queue?

*) Postfix evaluation requires LIFO order

*) The most recently added operands must be accessed first

*) A stack supports push and pop operations efficiently

*) A queue follows FIFO which does not match postfix evaluation requirements.